
(a)
ibm_berlin (RMS 0.044)
shot noise only (RMS 0.028)
FakeBerlin (RMS 0.029)
density
10^2
10^1
10^0
10^{-1}
−0.2
−0.1
0.0
0.1
0.2
$\langle Z\rangle_{\text{meas}} - \hat{f}(\boldsymbol{\theta})$ (calibration)
(b)
contrast $C_{\text{ideal}}^{(q)}$
1.000
0.975
0.950
0.925
0.900
0.875
0.850
ibm_berlin
FakeBerlin
1
4
7
10
output qubit q
(c)
dotted: 3 x median (drop)
calibration misfit (RMS)
0.10
0.08
0.06
0.04
0.02
shot noise
1
2
3
4
5
6
7
8
chain
(d)
ibm_berlin, training centres
FakeBerlin, training centres
ibm_berlin, validation (8-chain mean)
RMS of $\Delta = \langle Z\rangle_{\text{QPU}} - \hat{f}$
0.10
0.08
0.06
0.04
0.02
0.00
$1/\sqrt{S}$
0
25
50
75
100
epoch